Clinical Practice Guideline: Diabetes Mellitus Diagnosis & Management

1. Glycated Haemoglobin (HbA1c) Reference Standards

HbA1c reflects average plasma glucose over the preceding 8 to 12 weeks. Normal HbA1c is below 5.7%. Prediabetes is defined as HbA1c between 5.7% and 6.4%. An HbA1c of 6.5% or higher confirms diabetes mellitus. An HbA1c above 8.0% indicates poorly controlled diabetes with high risk of microvascular complications such as diabetic nephropathy and retinopathy.

2. Fasting Blood Sugar (FBS) and Post-Prandial Blood Sugar (PPBS)

Fasting Blood Sugar (FBS) is measured after an 8-hour fast. Normal FBS is 70 to 99 mg/dL. Impaired fasting glucose (prediabetes) is 100 to 125 mg/dL. FBS of 126 mg/dL or higher indicates diabetes. Post-Prandial Blood Sugar (PPBS) 2 hours after a meal is normal below 140 mg/dL, prediabetic from 140 to 199 mg/dL, and diabetic at 200 mg/dL or above.

3. Clinical Management & Lifestyle Protocol

Patients with elevated HbA1c (> 7.0%) require medical nutrition therapy (reducing refined carbohydrates and sugary drinks), 150 minutes of weekly aerobic activity, and pharmacological review (e.g., Metformin as first-line therapy if renal function eGFR > 45 mL/min).

Test Marker	Normal Range	Prediabetes	Diabetes Threshold
HbA1c (%)	< 5.7%	5.7% - 6.4%	>= 6.5%
FBS (mg/dL)	70 - 99 mg/dL	100 - 125 mg/dL	>= 126 mg/dL
2-Hr PPBS (mg/dL)	< 140 mg/dL	140 - 199 mg/dL	>= 200 mg/dL